

Name: Anthony Claudet

Date: _____

Class: _____

Element E: Application of STEM Principles and Practices

STEM Principles and Practices to Design Requirements (E1)

High-Priority Design Requirement #1 from Element C - The program must evaluate each device and recommend delaying, turning off, or keeping the device on based on whether it is peak energy hours and whether the device is essential.

STEM Concepts Involved in the requirement: Computing, math, functions, coding, technology, laws, reliability engineering

Describe how STEM concepts apply directly to the specifics of your design requirement:

The system uses Boolean logic to make decisions that mirror real-world energy-saving strategies used in smart grids.

High-Priority Design Requirement #2 from Element C - The program must allow user input to define each device's name, power usage (in kW), and whether it is essential or non-essential.

STEM Concepts Involved in the requirement: Programming, input/output, booleans, variables, mathematical units

Describe how STEM concepts apply directly to the specifics of your design requirement:



Name: Anthony Claudet**Date:** _____**Class:** _____

Involves if/else statements, applies real-world peak/off-peak energy patterns, determines system behavior based on feedback, and uses important units such as kW.

High-Priority Design Requirement #3 from Element C - The program must categorize time of day as either "peak" or "off-peak" hours based on user input and adjust recommendations accordingly.

STEM Concepts Involved in the requirement: Efficiency engineering, booleans, computing, data structures

Describe how STEM concepts apply directly to the specifics of your design requirement:

Uses arrays and lists to store device data, calculates energy saved, and ensures usability and proper data collection.

High-Priority Design Requirement #4 from Element C (if you have one)- The program must output clear recommendations for each device.

STEM Concepts Involved in the requirement: Computer interaction, information design, and decision systems

Describe how STEM concepts apply directly to the specifics of your design requirement:



Name: Anthony Claudet

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Class: _____

Focuses on how users receive and interpret system output, ensures clarity in presenting decisions, and provides actionable insights to help users make efficient choices.

High-Priority Design Requirement #5 from Element C (if you have one) - The program must be able to analyze at least five devices in one session without crashing or freezing.

STEM Concepts Involved in the requirement: Algorithm efficiency, memory management, reliability engineering

Describe how STEM concepts apply directly to the specifics of your design requirement:

These concepts are applied to keep the program stable and responsive, a key factor in user trust and system scalability.

Proposed Solution to STEM Principles and Practices (E2)

*STEM Principles for Design Solution #1 - **Smart Device List Evaluator (Text-Based Console Program)***

STEM Concept: Boolean logic, energy load awareness, data structures

Describe how STEM concept applies directly to the specifics of the first solution to your problem:

Makes energy-saving decisions by checking if a device is non-essential (booleans) and it's peak time. Applies knowledge of peak and off-peak electricity demand to simulate real-world utility patterns (energy load awareness). Uses arrays and lists to store device



Name: Anthony Claudet

Date: _____

Class: _____

information and for loops to evaluate it (data structures).



You can only work on the next sections after you have built and tested your first prototype. You need to skip to the “Evidence is Reviewed” section.

*STEM Principles for Design Solution #2 - **Visual Priority Dashboard (GUI-Based Representation)***

STEM Concept: User interface design, energy efficiency modeling, logical computing

Describe how STEM concept applies directly to the specifics of your solution:

Creates a highly interactive visual display featuring a UI (user interface). Visually represents device status in relation to energy use (energy efficiency modeling). Implements clickable elements and status updates to respond to user input (logical computing).

STEM Principles for Design Solution #3 (if completed)

STEM Concept: Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.

Describe how STEM concept applies directly to the specifics of your solution:



Name: Anthony Claudet

Date: _____

Class: _____

**Evidence Design is Reviewed (E3)***Documentation #1 (Only one documentation section needed for full credit):*

What format: notes

- Using Boolean logic to drive decision making is an appropriate and scalable foundation for this kind of simulation
- To increase accuracy, one improvement might be to factor in time duration, not just time category
- Good inclusion of control systems thinking. This mirrors how actual home energy managers work based on prioritization

Experts with Credentials (E4)

Expert #1 (from documentation 1): Product manager at Snap who graduated from Cambridge. Heavily involved in engineering projects and optimization.

Expert #2 (from documentation 2): Experienced software engineer who also works in game development. Codes in Unity and C#, often managing data storage and systems.

Expert #3 (from documentation 3): n/a

Verification or Corrective Response to Design Review (E5)

Expert #1 emphasized that the energy unit used (kW) could be translated to dollars saved and/or carbon emissions reduced, causing me to reconsider my units.



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Interview with Expert #1 informs corrective response: Recommend continuing the project and combining the design prototypes into something broader.

Interview with Expert #2 informs corrective response: For further development, consider real-time sensor input or historical usage patterns.

